

# Yash Rawal

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## Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms and ERP AI infrastructure across 120+ organizations. Specialized in core model training, embedding optimization, and low-latency agent execution frameworks. Active contributor to Spring AI, bridging machine learning models with high-throughput production systems.

## Technical Skills

**Languages & Backend:** Python, Java, SQL, Spring Boot, FastAPI, PostgreSQL, SAP HANA, Data Engineering  
**AI & ML Engineering:** Machine Learning, Model Training & SFT, Agentic AI, MCP, Vector Embeddings  
**AI & ML Frameworks:** PyTorch, Spring AI, LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, vLLM, SAP Build, SAP Joule  
**Platforms & Models:** Azure OpenAI, SAP AI Core, Amazon Bedrock, Google Vertex AI, Claude, Gemini, Qwen, Meta Llama & Open Weights  
**Infrastructure & Security:** Docker, Kubernetes, Cloud Foundry, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization (BIO NER)

## Experience

### TalenTeam Ltd., UK (via Bigscal Technologies)

**AI/ML Engineer** Oct 2024 - Present  
Remote

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by **70%**.
- Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned **E5-large-v2** embeddings using PyTorch to optimize query intent mapping.
- Developed the Prompt Lab evaluation system utilizing FastAPI and n8n webhooks to capture reasoning trajectories and human annotations for LLM-as-a-Judge grading.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Secured agentic workflows against OWASP LLM vulnerabilities via runtime tool monitoring and trained token-classification models for automated PII redaction.

### Healthray (Bigscal Technologies)

**AI/ML Engineer Intern** Apr 2024 - Sep 2024  
Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

## Open Source & Projects

### Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.  
Java, Spring Boot

### Modular Agent Harness & Memory System ("Brain") | Open Source

**Principal** architect of an open-source plug-and-play agentic framework designed as modular, composable building blocks for flexible enterprise workflows, implementing a formal Java-based ReAct loop (Thought-Action-Observation) with step budgets, tactical pgvector tool routing, and dynamic MCP settings caching.  
Java, Spring Boot, PostgreSQL, WebSockets

### AI Agent Security Red Teaming - OpenAI | Security Researcher

Designed adversarial prompt-injection pipelines targeting guardrails to evaluate security boundaries across 2,000+ agent test cases.  
Python, Kaggle

## Education

**Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University** 2022 - 2024

## Certifications

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|-----------------------------------------------------|-----------|
| Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft |
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